



Evaluation of Speed of Sound and Specific Heat Capacities of Real Gases

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In this paper, by use of the second virial coefficient (12-6) in the Lennard–Jones potential, an analytical procedure to evaluate the specific heat capacity and speed of sound of gases is presented. The obtained formulas yield accurate and fast evaluation of both quantities. By using obtained analytical expressions, the heat capacity and speed of sound of the gases Ar, He, Xe, H₂, O₂, F₂, Cl₂, NF₃, BCl₃, Ga(CH₃)₃, CO₂, and air were calculated in the temperature range from 100 to 2000 K and over the pressure range from 0.1 atm to approximately 15 atm. The computational results are in good agreement with the data available in the literature.

I. Introduction

STUDYING the thermodynamic and transport properties of gases is important for many industrial applications [1,2]. In addition, accurate evaluations of the thermodynamic and transport properties of the fluid are necessary for many chemical processes, to optimize equipment, in design of engineering systems, and in the semiconductor processing industry [2–5]. Note that evaluation of thermodynamic and transport properties of gases, liquids, and solids is still one of the main problems in science. It can be seen from the literature that these properties are widely investigated using both experimental and theoretical methods [6–13]. The thermodynamic properties of gases, liquids, and solids, necessary in the design and phase behavior calculations, are generally determined through the equations of state [14–19].

The equations of state are applicable to all kinds of gases within certain pressure and temperature ranges. It is well known that the virial equation of state, an alternative to the van der Waals equation of state, can be derived from statistical mechanical theory [20–25]. Note that the virial equation of state is a fundamental equation describing thermodynamic properties of gases at low to moderate densities [26,27]. The heat capacity and speed of sound are of fundamental importance in the study of various properties of gases. They are used in the engineering design of chemical processes. Examples include the generalization of heat exchanger and energy balance design calculations or in gas-phase chemical reactions. These data are essential for determining the energy necessary to bring the chemical reactants up to reaction temperature.

Because the virial equations of state define the various thermodynamic properties of real gases, it is necessary to derive the practical formulas from them regardless of whether they include differential and integral operations. In this study, we present a new method to evaluate the heat capacity at constant volume and pressure, and the speed of sound, of real gases. For this purpose, we have established analytical formulas for the first and second derivatives of the second virial coefficient. Finally, the established formulas are

used to calculate the heat capacity at constant volume and pressure as well as the speed of sound of the gases Ar, He, Xe, H₂, O₂, F₂, Cl₂, NF₃, BCl₃, Ga(CH₃)₃, CO₂, and air.

II. Theory

A. Definition of the Specific Heat Capacities and Speed of Sound

The thermodynamic functions of the real gases can be written in terms of the virial coefficients in the following forms [28].

For heat capacity at constant volume,

$$C_V - C_V^0 = -R \left(\frac{2T}{V} \frac{dB(T)}{dT} + \frac{T^2}{V} \frac{d^2B(T)}{dT^2} + \dots \right) \quad (1)$$

For heat capacity at constant pressure,

$$C_P - C_P^0 = -R \left(\frac{T^2}{V} \frac{d^2B(T)}{dT^2} - \frac{(B(T) - T(dB(T)/dT))^2}{V^2} + \dots \right) \quad (2)$$

For speed of sound,

$$c_0^2 = \frac{\gamma RT}{M} \left[1 + \frac{1}{V} \left(2B(T) + 2(\gamma - 1)T \frac{dB(T)}{dT} + \frac{(\gamma - 1)^2}{\gamma} T^2 \frac{d^2B(T)}{dT^2} \right) + \dots \right] \quad (3)$$

with

$$\gamma = C_P/C_V \quad (4)$$

$$\gamma = C_P/(C_P - R) \quad (5)$$

Here, $B(T)$ is the second virial coefficient; V is the volume; T is the temperature; R is the universal gas constant; M is the molecular weight; and γ is the heat capacity ratio. The superscript zero refers to the particular property of a gas in its ideal state [28].

B. Expressions for the First and Second Derivative of the Second Virial Coefficient

As can be seen from Eqs. (1–3), the choice of reliable formulas for the second virial coefficient is of prime importance for accurate calculations of the specific heat capacity and speed of sound of real

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Table 1 Lennard–Jones (12-6) potential parameters [31,39,40]

Nonpolar molecules	σ , Å	b_0 , cal/mol · atm	ϵ/k_B , K	M , g/mol
H ₂	2.920	0.760341	37.020	2.016000
O ₂	3.580	1.401230	117.50	31.99980
Cl ₂	4.115	2.127990	357.00	70.90600
CO ₂	3.967	1.906540	208.10	44.01000
Ar	3.410	1.210940	119.50	39.94800
He	2.630	0.556000	6.0300	4.002 602
Xe	4.050	2.028730	230.00	131.2930
Air	3.620	1.448720	97.000	28.96600
F ₂	3.357	1.155350	112.60	37.99600
NF ₃	4.280	2.394370	173.86	71.00000
BCl ₃	5.006	3.831180	346.80	117.1700
Ga(CH ₃) ₃	5.470	4.998290	378.20	114.8266

gases. We therefore use the following form for the second virial coefficient [29]:

$$B(T) = \frac{2\pi N_A \sigma^3}{3} \frac{8\epsilon}{\sqrt{2} k_B T} \lim_{N \rightarrow \infty} \sum_{k=0}^N \frac{(2\epsilon/k_B T)^k}{k!} \times \left(\frac{1}{4} (\epsilon/k_B T)^{-(3/4)-(k/2)} \Gamma\left(\frac{3}{4} + \frac{k}{2}\right) - \frac{1}{4} (\epsilon/k_B T)^{-(1/4)-(k/2)} \Gamma\left(\frac{1}{4} + \frac{k}{2}\right) \right) \quad (6)$$

where $b_0 = 2\pi N_A \sigma^3/3$; N_A is Avogadro's constant; k_B is Boltzmann's constant; σ is the finite distance at which the interparticle potential is zero; and ϵ is the depth of the potential well. In Eq. (6), the index N is the upper limit of summation. The first and second derivatives of second virial coefficient can be written as, respectively,

$$B_1(T) = T \frac{dB(T)}{dT} \quad (7)$$

$$B_2(T) = T^2 \frac{d^2 B(T)}{dT^2} \quad (8)$$

Taking the first and second derivatives of Eq. (6) with respect to T , we find

$$B_1(T) = \frac{2\pi N_A \sigma^3}{3} T \lim_{N \rightarrow \infty} \sum_{k=0}^N \frac{2^{k+1}}{\sqrt{2} k!} \times \left(\left(-\frac{1}{4} - \frac{k}{2} \right) (\epsilon/k_B)^{(1/4)+(k/2)} T^{-(5/4)-(k/2)} \Gamma\left(\frac{3}{4} + \frac{k}{2}\right) - \left(-\frac{3}{4} - \frac{k}{2} \right) (\epsilon/k_B)^{(3/4)+(k/2)} T^{-(7/4)-(k/2)} \Gamma\left(\frac{1}{4} + \frac{k}{2}\right) \right) \quad (9)$$

Table 2 Speed of sound for He

T , K	P , atm	u , m · s ⁻¹	u , m · s ⁻¹ (experiment) [34]	u , m · s ⁻¹ [44]
400.00	14.8762	1182.21	1182.215	1176.8
	13.9909	1181.91	1181.882	1176.8
	12.4324	1181.38	1181.327	1176.8
	10.4189	1180.69	1180.618	1176.8
	9.26454	1180.30	1180.227	1176.8
	7.33748	1179.64	1179.550	1176.8
	5.86677	1179.14	1179.051	1176.8
350.00	14.9676	1106.72	1106.812	1100.8
	13.4090	1106.14	1106.193	1100.8
	12.0138	1105.61	1105.616	1100.8
	10.1959	1104.93	1104.928	1100.8
	9.07002	1104.51	1104.515	1100.8
	7.25290	1103.83	1103.819	1100.8
	5.81762	1103.29	1103.238	1100.8
	3.90220	1102.57	1102.563	1100.8
	2.00089	1101.85	1101.872	1100.8
300.01	14.1331	1025.34	1025.584	1019.3
	12.1506	1024.52	1024.720	1019.3
	10.4528	1023.81	1023.979	1019.3
	9.00291	1023.20	1023.291	1019.3
	7.44002	1022.55	1022.589	1019.3
	5.59013	1021.78	1021.817	1019.3
	3.52647	1020.92	1020.840	1019.3
	2.53541	1020.50	1020.506	1019.3
	1.50693	1020.07	1020.047	1019.3
250.00	14.8515	937.632	938.097	930.3
	13.6986	937.088	937.493	930.3
	12.1295	936.348	936.701	930.3
	10.3736	935.519	935.816	930.3
	8.53817	934.651	934.881	930.3
	6.56304	933.716	933.909	930.3
	5.08236	933.014	933.121	930.3
	3.47081	932.249	932.288	930.3
	2.00829	931.555	931.607	930.3
225.00	14.9012	890.425	890.929	882.6
	13.8438	889.890	890.338	882.6
	12.8561	889.390	889.751	882.6
	11.9425	888.926	889.273	882.6
	10.3581	888.122	888.414	882.6
	9.62783	887.752	888.033	882.6
	8.40839	887.132	887.345	882.6

Table 3 Speed of sound for Xe

T , K	P , atm	u , m · s ⁻¹	u , m · s ⁻¹ (experiment) [34]	u , m · s ⁻¹ [44]
399.97	13.46770	204.221	204.1431	205.5
	11.68550	204.396	204.3155	205.5
	9.632690	204.598	204.5105	205.5
	7.997280	204.758	204.6676	205.5
	6.678360	204.888	204.8020	205.5
	4.859650	205.066	204.9795	205.5
	3.784150	205.172	205.0859	205.5
	2.655810	205.282	205.2015	205.5
	1.432190	205.402	205.3316	205.5
	0.552874	205.488	205.4201	205.5
359.98	14.30610	192.523	192.4631	194.9
	11.09430	193.081	193.0096	194.9
	8.975310	193.448	193.3733	194.9
	6.206840	193.927	193.8524	194.9
	4.144650	194.283	194.2122	194.9
	2.677940	194.536	194.4691	194.9
	1.810910	194.685	194.6235	194.9
299.99	11.89980	173.807	173.5890	177.9
	8.937480	174.862	174.7120	177.9
	5.010810	176.251	176.1640	177.9
	2.060500	177.288	177.2350	177.9
	0.578633	177.806	177.7650	177.9
260.00	9.070520	160.672	160.2140	165.7
	5.972270	162.414	162.1760	165.7
	3.091340	164.017	163.9060	165.7
	1.020680	165.159	165.1030	165.7
	0.709006	165.331	165.8100	165.7
	0.405823	165.497	165.4550	165.7
209.97	1.952330	146.966	146.7770	148.9
	0.800691	148.124	148.0360	148.9
	0.657192	148.268	148.1900	148.9
	0.539946	148.385	148.3140	148.9
	0.405033	148.520	148.4580	148.9
	0.404935	148.520	148.4570	148.9

$$B_2(T) = \frac{2\pi N_A \sigma^3}{3} T^2 \lim_{N \rightarrow \infty} \sum_{k=0}^N \frac{2^{k+1}}{\sqrt{2k!}} \left(\left(-\frac{1}{4} - \frac{k}{2} \right) \left(-\frac{5}{4} - \frac{k}{2} \right) \right. \\ \times (\varepsilon/k_B)^{(1/4)+(k/2)} T^{-(9/4)-(k/2)} \Gamma\left(\frac{3}{4} + \frac{k}{2}\right) - \left(-\frac{3}{4} - \frac{k}{2} \right) \\ \times \left(-\frac{7}{4} - \frac{k}{2} \right) (\varepsilon/k_B)^{(3/4)+(k/2)} T^{-(11/4)-(k/2)} \Gamma\left(\frac{1}{4} + \frac{k}{2}\right) \left. \right) \quad (10)$$

The quantities $\Gamma(\alpha)$ in Eqs. (6), (9), and (10), are the well-known gamma functions defined in [30]:

$$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt \quad (11)$$

Equations (1–3) can be written in terms of the quantities $B(T)$, $B_1(T)$, and $B_2(T)$ to produce the following.

For heat capacity at constant volume,

$$C_V = C_V^0 - R \left(\frac{2}{V} B_1(T) + \frac{1}{V} B_2(T) \right) \quad (12)$$

For heat capacity at constant pressure,

$$C_P = C_P^0 - R \left(\frac{B_2(T)}{V} + \frac{(B(T) - B_1(T))^2}{V^2} \right) \quad (13)$$

For the speed of sound,

$$c_0^2 = \frac{\gamma RT}{M} \left[1 + \frac{1}{V} \left(2B(T) + 2(\gamma - 1)B_1(T) \right. \right. \\ \left. \left. + \frac{(\gamma - 1)^2}{\gamma} B_2(T) \right) + \dots \right] \quad (14)$$

III. Numerical Results and Discussion

In this paper, we propose an efficient approach to calculating the heat capacity at constant volume, heat capacity at constant pressure, and the speed of sound, virial coefficient with the Lennard–Jones (12–6) potential. The obtained formulas are completely general and free of any restrictions in terms of applications. To demonstrate the accuracy and efficiency of the analytical expressions, we present several calculations of properties for the gases Ar, He, Xe, H₂, O₂, F₂, Cl₂, NF₃, BCl₃, Ga(CH₃)₃, CO₂, and air. Based on Eqs. (12–14), we constructed programs for computation of the heat capacity at constant volume, heat capacity at constant pressure, and the speed of sound, using Pascal 7.0 international mathematical software.

The heat capacities and speed of sound were determined in the temperature range from 100–2000 K and over the pressure range from 0.1 atm to approximately 15 atm. The calculated results were compared with experimental and theoretical data from the literature. The parameters σ , b_0 , ε/k_B , and M are used to describe some molecules in Table 1. In Tables 2–19, the calculated results are compared with the corresponding experimental and other theoretical

Table 4 Speed of sound for Ar

P , atm	T , K	u , m · s ⁻¹	u , m · s ⁻¹ [33]	u , m · s ⁻¹ [44]
0.789539	120.0	204.315	203	204.0
	150.0	228.470	228	228.1
	200.0	263.786	263	263.4
	300.0	322.976	323	322.6
	400.0	372.866	373	372.5
	500.0	416.821	417	416.5
	600.0	456.561	456	456.2
	700.0	493.108	493	492.8
	800.0	527.127	527	526.8
	900.0	559.079	559	558.7
	1000	589.301	589	589.0
1.000000	120.0	204.374	203	204.0
	150.0	228.549	227	228.1
	200.0	263.870	263	263.4
	300.0	323.053	323	322.6
	400.0	372.936	373	372.5
	500.0	416.884	417	416.5
	600.0	456.619	456	456.2
	700.0	493.161	493	492.8
	800.0	527.176	527	526.8
	900.0	559.125	559	558.7
	1000	589.345	589	589.0
9.869230	120.0	205.657	187	204.0
	150.0	231.747	222	228.1
	200.0	267.481	262	263.4
	300.0	326.365	323	322.6
	400.0	375.894	374	372.5
	500.0	419.553	418	416.5
	600.0	459.055	458	456.2
	700.0	495.407	494	492.8
	800.0	529.263	528	526.8
	900.0	561.078	560	558.7
	1000	591.182	590	589.0
14.80380	150.0	233.352	218	228.1
	200.0	269.531	261	263.4
	300.0	328.233	324	322.6
	400.0	377.549	375	372.5
	500.0	421.042	419	416.5
	600.0	460.412	459	456.2
	700.0	496.656	495	492.8
	800.0	530.423	529	526.8
	900.0	562.163	561	558.7
	1000	592.202	591	589.0

Table 5 Speed of sound for O₂

P , atm	T , K	u , m · s ⁻¹	u , m · s ⁻¹ [35]	u , m · s ⁻¹ [44]
0.197385	100	190.639	190	198.1
	120	208.918	208	211.8
	140	225.704	225	226.8
	160	241.317	241	241.6
	180	255.973	255	256.0
	200	269.831	269	269.7
	220	283.008	282	282.8
	240	295.597	295	295.3
	260	307.670	307	307.3
	300	330.492	329	329.7
1.000000	100	190.064	188	198.1
	120	208.636	207	211.8
	140	225.593	224	226.8
	160	241.313	240	241.6
	180	256.041	255	256.0
	200	269.948	269	269.7
	220	283.160	282	282.8
	240	295.774	295	295.3
	260	307.865	307	307.3
	300	330.710	329	329.7
9.86923	140	224.676	214	226.8
	160	241.572	233	241.6
	180	257.037	250	256.0
	200	271.437	266	269.7
	220	284.992	280	282.8
	240	297.848	293	295.3
	260	310.113	306	307.3
	300	333.173	329	329.7
14.8038	140	224.311	207	226.8
	160	241.909	229	241.6
	180	257.763	248	256.0
	200	272.406	264	269.7
	220	286.122	279	282.8
	240	299.090	293	295.3
	260	311.434	306	307.3
	300	334.587	330	329.7

Table 6 Speed of sound for F₂

<i>P</i> , atm	<i>T</i> , K	<i>u</i> , m · s ⁻¹	<i>u</i> , m · s ⁻¹ [40]	<i>u</i> , m · s ⁻¹ [44]
0.10000	100	174.813	173	171.4
	120	191.767	191	189.8
	140	207.149	206	206.1
	160	221.462	221	220.7
	180	234.902	234	234.2
	200	247.612	246	246.6
	220	259.701	258	258.2
	240	271.250	269	296.1
	260	282.327	279	279.5
	300	303.268	299	298.9
0.493462	120	191.680	186	189.8
	140	207.121	203	206.1
	160	221.471	219	220.7
	180	234.936	233	234.2
	200	247.663	246	246.6
	220	259.763	257	258.2
	240	271.321	269	269.1
	260	282.403	279	279.5
	300	303.352	299	298.9
0.986923	120	191.572	179	189.8
	140	207.086	199	206.1
	160	221.483	216	220.7
	180	234.978	231	234.2
	200	247.727	245	246.6
	220	259.841	257	258.2
	240	271.41	268	269.1
	260	282.50	279	279.5
	300	303.458	299	298.9
1.38169	120	191.486	172	189.8
	140	207.059	196	206.1
	160	221.493	214	220.7
	180	235.013	230	234.2
	200	247.778	244	246.6
	220	259.905	256	258.2
	240	271.481	268	269.1
	260	282.577	279	279.5
	300	303.542	299	298.9

Table 7 Speed of sound for H₂

<i>P</i> , atm	<i>T</i> , K	<i>u</i> , m · s ⁻¹	<i>u</i> , m · s ⁻¹ [48]	<i>u</i> , m · s ⁻¹ [44]
0.394769	100	760.502	772.30	
	150	931.346	911.30	
	200	1075.33	1054.0	
	250	1202.18	1188.0	
	300	1316.86	1309.0	
	360	1442.48	1439.0	
	400	1520.47	1518.0	
1.000000	100	761.154	772.70	775.8
	150	932.027	911.90	952.8
	200	1075.97	1054.0	1087.9
	250	1202.77	1189.0	1207.8
	300	1317.41	1310.0	1318.5
	360	1442.99	1439.0	1441.6
	400	1520.95	1518.0	1518.9
9.86923	100	770.999	779.20	775.8
	150	942.052	919.80	952.8
	200	1085.37	1063.0	1087.9
	250	1211.47	1197.0	1207.8
	300	1325.47	1317.0	1318.5
	360	1450.40	1446.0	1441.6
	400	1527.99	1525.0	1518.9
14.8038	100	776.713	783.20	775.8
	150	947.675	924.40	952.8
	200	1090.60	1067.0	1087.9
	250	1216.30	1201.0	1207.8
	300	1329.95	1322.0	1318.5
	360	1454.52	1450.0	1441.6
	400	1531.90	1529.0	1518.9
23.6862	100	787.393	791.40	775.8
	150	957.877	932.80	952.8
	200	1100.02	1079.0	1087.9
	250	1224.99	1209.0	1207.8
	300	1338.00	1329.0	1318.5
	360	1461.91	1457.0	1441.6
	400	1538.92	1535.0	1518.9

values [5,31–44], and satisfactory consistency is presented. The correctness of the proposed method is acceptable and can be suggested for evaluation of specific heat capacities and speed of sound of real gases. The quantities ΔC_P and ΔC_V correspond to $C_P - C_P^0$ and $C_V - C_V^0$, respectively. Also, the results for molecular systems show a good rate of convergence and numerical stability. It is seen from the algorithm presented for quantities $B(T)$, $B_1(T)$, and $B_2(T)$ that our CPU times are satisfactory. As an example, in the case of the second virial coefficient with upper limit $N = 20$, the CPU times to use Eq. (6) [28] and at temperature $T = 400$ K are about 0.002 and 0.015 ms, respectively. As can be seen from our obtained results with different molecules, the proposed algorithms are general and significant in the evaluation of various properties of gases for a wide range of values of thermodynamic parameters. All calculations were performed with $N = 50$. As a tool, this calculation method is conceptually valid over wide temperature ranges, and it offers some advantages over currently available methods [45–47]. The calculated results of speed of sound for He, NF₃, BCl₃, CO₂, Cl₂, and Xe are compared with experimental and theoretical data [32,34,38,41], and results for Ar, O₂, F₂, H₂, and air are compared with theoretical data [33,35,37,40,44,48]. The calculated results of speed of sound of Ga(CH₃)₃ are compared with experimental data [41]. The calculated results of the heat capacities at constant pressure for Ar, O₂, He, Xe, NF₃, BCl₃, CO₂, Cl₂, F₂, H₂ and air are compared with theoretical data [33,35,37,40,44,48]. The calculated results of heat capacities at constant volume for Ar, O₂, He, Xe, F₂, H₂, and air are compared with theoretical data [33,35–37,40,48,49]. The results of calculation of speed of sound and specific heat capacities for He are in good agreement in the wide range of parameters. The specific heat capacities for Ar obtained from Eqs. (12) and (13) are in good agreement with [33,44] in the temperature range from 100 to 1000 K

and the pressure range from 0.8 to 15 atm. The obtained results for the specific heat capacities for O₂, CO₂, F₂, and H₂ have satisfactory values with the available data in [44]. The calculation results of C_V for O₂ agree with [35] in the temperature range from 100 to 300 K. The obtained results of C_P for O₂ have approximate values with [35,44] in the temperature range from 100 to 300 K and agree with [44] values of C_P better than [35]. The obtained results of C_P for CO₂, Xe, and air have approximate values with [44]. The C_V values obtained from Eq. (13) for Xe agree with [49]. As the temperature of gas increases, C_P of Cl₂ and air and C_V of air deviate from literature values [44].

The results of C_P for NF₃ and Cl₂ have approximate values with [44] in the temperature range from 198.98 to 260 K and the pressure from 10 to 0.9 atm and in the temperature range from 260 to 440 K and the pressure from 0.9 to 15 atm, respectively.

The calculation results of the speed of sound are compared with the available data in [32,33,37,40,41,43,44,48], and we obtained approximate values for H₂, Ar, F₂, NF₃, BCl₃, Cl₂, and air. The speed of sound values for O₂, NF₃, Cl₂, and CO₂ obtained from Eq. (14) agree with [44] results better than experimental and theoretical data [35,38,41,43].

The results of the speed of sound for Ga(CH₃)₃ have the approximate values of [41] in the temperature range from 340 to 420 K and the pressure range of from 0.3 to 8.4 atm. As can be seen in Table 13, the proposed method is significantly less accurate for C_P of BCl₃. The difference between our values and [44] of C_P is an average of 40% for BCl₃. The calculated results of the speed of sound for Ga(CH₃)₃ are not agreed upon in the literature. The difference between our values and [41] of the speed of sound is an average of 10% for Ga(CH₃)₃. This method seems to determine the speed of sound more accurately for the gases He, Ar, and Xe than the other gases. The heat capacity ratios have been determined from the calculated values of C_V and C_P for real gases. These ratios have been

Table 8 Specific heat capacities for Ar

P , atm	T , K	Density, mol/dm ³ [33]	C_P , J/(mol · K)	C_P , J/(mol · K) [33]	C_P , J/(mol · K) [44]	C_V , J/(mol · K)	C_V , J/(mol · K) [33]	ΔC_P , ^a J/(mol · K)	ΔC_V , ^a J/(mol · K)
0.789539	120.0	0.08104	21.1662	21.33	20.7849	12.5592	12.67	3.81725E – 1	8.84528E – 2
	150.0	0.06450	20.9909	21.02	20.7849	12.5106	12.53	2.06367E – 1	3.98960E – 2
	200.0	0.04822	20.8829	20.89	20.7849	12.4862	12.49	9.84182E – 2	1.54824E – 2
	300.0	0.03209	20.8216	20.82	20.7849	12.4755	12.48	3.70772E – 2	4.83570E – 3
	400.0	0.02406	20.8036	20.80	20.7849	12.4731	12.47	1.91021E – 2	2.43220E – 3
	500.0	0.01924	20.7960	20.80	20.7849	12.4723	12.47	1.15271E – 2	1.54710E – 3
	600.0	0.01603	20.7922	20.79	20.7849	12.4718	12.47	7.65195E – 3	1.12016E – 3
	700.0	0.01374	20.7899	20.79	20.7849	12.4716	12.47	5.41373E – 3	8.76211E – 4
	800.0	0.01203	20.7885	20.79	20.7849	12.4714	12.47	4.00865E – 3	7.20061E – 4
	900.0	0.01069	20.7876	20.79	20.7849	12.4713	12.47	3.07131E – 3	6.11827E – 4
	1000	0.00962	20.7869	20.79	20.7849	12.4712	12.47	2.41632E – 3	5.32345E – 4
1.000000	120.0	0.10295	21.2680	21.48	20.7849	12.5827	12.72	4.83478E – 1	1.12031E – 1
	150.0	0.08182	21.0459	21.08	20.7849	12.5212	12.54	2.61377E – 1	5.05308E – 2
	200.0	0.06269	20.9092	20.92	20.7849	12.4903	12.50	1.24653E – 1	1.96094E – 2
	300.0	0.04065	20.8315	20.83	20.7849	12.4768	12.48	4.69606E – 2	6.12471E – 3
	400.0	0.03047	20.8087	20.81	20.7849	12.4738	12.48	2.41940E – 2	3.08053E – 3
	500.0	0.02437	20.7991	20.80	20.7849	12.4727	12.47	1.45998E – 2	1.95950E – 3
	600.0	0.02031	20.7942	20.79	20.7849	12.4721	12.47	9.69167E – 3	1.41875E – 3
	700.0	0.01741	20.7914	20.79	20.7849	12.4718	12.47	6.85682E – 3	1.10978E – 3
	800.0	0.01523	20.7896	20.79	20.7849	12.4716	12.47	5.07721E – 3	9.12003E – 4
	900.0	0.01354	20.7884	20.79	20.7849	12.4715	12.47	3.89000E – 3	7.74917E – 4
	1000	0.01218	20.7876	20.79	20.7849	12.4714	12.47	3.06042E – 3	6.74248E – 4
9.869230	150.0	0.86478	23.3641	24.33	20.7849	12.9694	13.24	2.57959E – 0	4.98700E – 1
	200.0	0.61928	22.0147	22.21	20.7849	12.6642	12.74	1.23023E – 0	1.93530E – 1
	300.0	0.40330	21.2480	21.27	20.7849	12.5312	12.53	4.63465E – 1	6.04462E – 2
	400.0	0.30077	21.0233	21.02	20.7849	12.5011	12.51	2.38776E – 1	3.04025E – 2
	500.0	0.24017	20.9286	20.92	20.7849	12.4901	12.50	1.44089E – 1	1.93388E – 2
	600.0	0.20002	20.8802	20.88	20.7849	12.4847	12.49	9.56494E – 2	1.40020E – 2
	700.0	0.17141	20.8522	20.85	20.7849	12.4817	12.48	6.76716E – 2	1.09526E – 2
	800.0	0.14998	20.8346	20.83	20.7849	12.4797	12.48	5.01082E – 2	9.00077E – 3
	900.0	0.13333	20.8229	20.82	20.7849	12.4784	12.48	3.83913E – 2	7.64783E – 3
	1000	0.12001	20.8147	20.81	20.7849	12.4774	12.48	3.02040E – 2	6.65431E – 3
14.80380	150.0	1.3553	24.6539	26.81	20.7849	13.2188	13.71	3.86938E – 0	7.48051E – 1
	200.0	0.94315	22.6299	23.00	20.7849	12.7610	12.87	1.84534E – 0	2.90294E – 1
	300.0	0.606668	21.4797	21.52	20.7849	12.5614	12.60	6.95198E – 1	9.06693E – 2
	400.0	0.45118	21.1427	21.14	20.7849	12.5163	12.53	3.58164E – 1	4.56037E – 2
	500.0	0.35996	21.0007	20.99	20.7849	12.4997	12.51	2.16134E – 1	2.90082E – 2
	600.0	0.29969	20.9280	20.92	20.7849	12.4917	12.50	1.43474E – 1	2.10029E – 2
	700.0	0.25680	20.8860	20.88	20.7849	12.4871	12.49	1.01507E – 1	1.64290E – 2
	800.0	0.22470	20.8597	20.85	20.7849	12.4842	12.49	7.51623E – 2	1.35011E – 2
	900.0	0.19975	20.8421	20.84	20.7849	12.4822	12.48	5.75870E – 2	1.14717E – 2
	1000	0.17981	20.8298	20.83	20.7849	12.4807	12.48	4.53060E – 2	9.98147E – 3

^aThe quantities ΔC_P and ΔC_V correspond to $C_P - C_P^0$ and $C_V - C_V^0$, respectively.

Table 9 Specific heat capacities for O₂

P , atm	T , K	Density, mol/dm ³ [35]	C_P , J/(mol · K)	C_P , J/(mol · K) [35]	C_P , J/(mol · K) [44]	C_V , J/(mol · K)	C_V , J/(mol · K) [35]	ΔC_P , ^a J/(mol · K)	ΔC_V , ^a J/(mol · K)
0.197385	100	0.02167	29.2810	29.25	24.620	20.8332	20.80	1.82674E – 1	4.86440E – 2
	120	0.02010	29.2054	29.22	27.2823	20.8090	20.83	1.07039E – 1	2.44560E – 2
	140	0.01721	29.1682	29.19	28.4023	20.7986	20.82	6.9838E – 2	1.40690E – 2
	160	0.01505	29.1473	29.16	28.8695	20.7934	20.81	4.89802E – 2	8.89624E – 3
	180	0.01337	29.1345	29.15	29.0551	20.7906	20.81	3.61720E – 2	6.03447E – 3
	200	0.01203	29.1261	29.15	29.1255	20.7888	20.82	2.77650E – 2	4.32162E – 3
	220	0.01094	29.1203	29.17	29.1575	20.7878	20.84	2.19591E – 2	3.23173E – 3
	240	0.01003	29.1161	29.20	29.1895	20.7870	20.87	1.77861E – 2	2.50336E – 3
	260	0.00925	29.1130	29.24	29.2343	20.7865	20.92	1.46884E – 2	1.99660E – 3
	300	0.00802	29.1088	29.39	29.3879	20.7859	21.07	1.04867E – 2	1.36199E – 3
1.000000	100	0.12484	30.0238	29.94	24.620	21.0310	20.89	9.25473E – 1	2.46446E – 1
	120	0.10301	29.6406	29.70	27.2823	20.9084	20.94	5.42286E – 1	1.23904E – 1
	140	0.08784	29.4521	29.50	28.4023	20.8558	20.90	3.53817E – 1	7.12778E – 2
	160	0.07663	29.3465	29.38	28.8695	20.8296	20.86	2.48146E – 1	4.50706E – 2
	180	0.06799	29.2816	29.31	29.0551	20.8151	20.84	1.83256E – 1	3.05721E – 2
	200	0.06112	29.2390	29.27	29.1255	20.8064	20.84	1.40664E – 1	2.18944E – 2
	220	0.05552	29.2096	29.26	29.1575	20.8009	20.85	1.11250E – 1	1.63727E – 2
	240	0.05086	29.1884	29.27	29.1895	20.7972	20.88	9.01087E – 2	1.26826E – 2
	260	0.04693	29.1727	29.30	29.2343	20.7946	20.93	7.44149E – 2	1.01153E – 2
	300	0.04065	29.1515	29.44	29.3879	20.7914	21.08	5.31282E – 2	6.90018E – 3
9.86923	140	0.94974	32.5902	34.09	28.4023	21.4880	21.92	5.35194E – 0	1.22284E – 0
	160	0.80153	31.5473	32.16	28.8695	21.2293	21.47	3.49190E – 0	7.03458E – 1
	180	0.69794	30.9069	31.18	29.0551	21.0862	21.22	2.44901E – 0	4.44812E – 1
	200	0.62009	30.4866	30.61	29.1255	21.0006	21.09	1.80860E – 0	3.01723E – 1
	220	0.55887	30.1963	30.28	29.1575	20.9461	21.03	1.38825E – 0	2.16081E – 1
	240	0.50924	29.9876	30.07	29.1895	20.9097	21.01	1.09796E – 0	1.61586E – 1
	260	0.46804	29.8327	29.95	29.2343	20.8843	21.03	8.89303E – 1	1.25168E – 1
14.8038	300	0.40337	29.6227	29.89	29.3879	20.8526	21.15	7.34418E – 1	9.98301E – 2
	140	1.51680	34.3362	38.14	28.4023	21.8397	22.67	5.24335E – 1	6.80994E – 2
	160	1.24660	32.7718	34.17	28.8695	21.4517	21.85	6.41508E – 0	1.37211E – 0
	180	1.07170	31.8112	32.40	29.0551	21.2371	21.44	5.23785E – 0	1.05519E – 0
	200	0.94507	31.1807	31.45	29.1255	21.1086	21.23	3.67351E – 0	6.67218E – 1
	220	0.84779	30.7453	30.89	29.1575	21.0269	21.13	2.71290E – 0	4.52585E – 1
	240	0.77003	30.4323	30.54	29.1895	20.9723	21.09	2.08237E – 0	3.24122E – 1
	260	0.70615	30.2000	30.32	29.2343	20.9343	21.09	1.64693E – 0	2.42380E – 1
	300	0.60685	29.8848	30.14	29.3879	20.8867	21.09	1.33395E – 0	1.87752E – 1

^aThe quantities ΔC_P and ΔC_V correspond to $C_P - C_P^0$ and $C_V - C_V^0$, respectively.

Table 10 Specific heat capacities for He

T , K	P , atm	C_P , J/(mol · K)	C_P , J/(mol · K) [44]	C_V , J/(mol · K)	ΔC_P , ^a J/(mol · K)	ΔC_V , ^a J/(mol · K)	P , atm	T , K	C_V , J/(mol · K)	C_V , J/(mol · K) [36]
400.00	14.8762	20.7819	20.7884	12.4789	$-2.65230\text{E}-3$	$8.22149\text{E}-3$	0.0986923	50	12.4713	12.4721
	12.4324	20.7823	20.7884	12.4776	$-2.21659\text{E}-3$	$6.87088\text{E}-3$		100	12.4710	12.4721
	9.26454	20.7829	20.7884	12.4758	$-1.65179\text{E}-3$	$5.12015\text{E}-3$		500	12.4707	12.4721
	7.33748	20.7832	20.7884	12.4748	$-1.30821\text{E}-3$	$4.05514\text{E}-3$		800	12.4707	12.4721
	5.86677	20.7835	20.7884	12.4740	$-1.10460\text{E}-3$	$3.24233\text{E}-3$		1000	12.4707	12.4712
350.00	14.9676	20.7826	20.7884	12.4804	$-1.90117\text{E}-3$	$9.65882\text{E}-3$	49.3462	1200	12.4707	12.4721
	12.0138	20.7830	20.7884	12.4785	$-1.52599\text{E}-3$	$7.75271\text{E}-3$		1400	12.4707	12.4721
	9.07002	20.7834	20.7884	12.4766	$-1.15207\text{E}-3$	$5.85303\text{E}-3$		1500	12.4707	12.4721
	7.25290	20.7836	20.7884	12.4754	$-9.21259\text{E}-4$	$4.68041\text{E}-3$		50	12.7771	12.7203
	5.81762	20.7838	20.7884	12.4745	$-7.38951\text{E}-4$	$3.75420\text{E}-3$		100	12.6040	12.6162
300.01	2.00089	20.7843	20.7884	12.4720	$-2.54152\text{E}-4$	$1.29121\text{E}-3$	493.462	500	12.4917	12.5041
	14.1331	20.7841	20.7884	12.4816	$-3.70136\text{E}-4$	$1.08983\text{E}-2$		800	12.4828	12.4961
	12.1506	20.7842	20.7884	12.4801	$-3.18215\text{E}-4$	$9.36950\text{E}-3$		1000	12.4800	12.4921
	9.00291	20.7843	20.7884	12.4777	$-2.35779\text{E}-4$	$6.94227\text{E}-3$		1200	12.4782	12.4921
	7.44002	20.7843	20.7884	12.4764	$-1.94848\text{E}-4$	$5.73710\text{E}-3$		1400	12.4769	12.4921
250.00	3.52647	20.7844	20.7884	12.4734	$-9.23557\text{E}-5$	$2.71931\text{E}-3$	986.923	1500	12.4764	12.4921
	1.50693	20.7845	20.7884	12.4719	$-3.94654\text{E}-5$	$1.16202\text{E}-3$		100	13.8036	13.6849
	14.8515	20.7872	20.7884	12.4848	$2.64335\text{E}-3$	$1.41235\text{E}-2$		300	12.8512	12.6162
	12.1295	20.7867	20.7884	12.4822	$2.15887\text{E}-3$	$1.15349\text{E}-2$		500	12.6809	12.5041
	8.53817	20.7860	20.7884	12.4788	$1.51967\text{E}-3$	$8.11961\text{E}-3$		800	12.5915	12.4961
225.00	6.56304	20.7857	20.7884	12.4770	$1.16812\text{E}-3$	$6.24130\text{E}-3$	986.923	1000	12.5633	12.4921
	5.08236	20.7854	20.7884	12.4755	$9.04583\text{E}-4$	$4.83321\text{E}-3$		1200	12.5452	12.4921
	2.00829	20.7849	20.7884	12.4726	$3.57445\text{E}-4$	$1.90984\text{E}-3$		1500	12.5277	12.4921
	14.9012	20.7898	20.7884	12.4867	$5.29261\text{E}-3$	$1.59875\text{E}-2$		150	14.1508	14.3653
	12.8561	20.7891	20.7884	12.4845	$4.56623\text{E}-3$	$1.37933\text{E}-2$		500	12.8911	13.2286
	11.9425	20.7888	20.7884	12.4835	$4.24174\text{E}-3$	$1.28131\text{E}-2$		1000	12.6559	12.9964
	10.3581	20.7882	20.7884	12.4818	$3.67899\text{E}-3$	$1.11132\text{E}-2$		1200	12.6197	12.8764
	8.40839	20.7875	20.7884	12.4797	$2.98650\text{E}-3$	$9.02141\text{E}-3$		1400	12.5945	12.8403
								1500	12.5847	12.8283

^aThe quantities ΔC_P and ΔC_V correspond to $C_P - C_P^0$ and $C_V - C_V^0$, respectively.

Table 11 Specific heat capacities for Xe

T , K	P , atm	C_P , J/(mol · K)	C_P , J/(mol · K) [44]	C_V , J/(mol · K)	ΔC_P , ^a J/(mol · K)	ΔC_V , ^a J/(mol · K)	P , atm	T , K	C_V , J/(mol · K)	C_V , J/(mol · K) [49]
399.97	13.46770	22.1132	20.7837	12.6748	1.32872E-00	2.04105E-01	0.986923	165	12.8258	13.3000
	9.632690	21.7349	20.7837	12.6167	9.50360E-01	1.459850E-01		300	12.5088	12.8011
	6.678360	21.4434	20.7837	12.5719	6.58887E-01	1.01212E-01		500	12.4785	12.4728
	2.655810	21.0465	20.7837	12.5110	2.62022E-01	4.02492E-02		600	12.4755	12.4728
	1.432190	20.9258	20.7837	12.4924	1.41300E-01	2.17051E-02		800	12.4732	12.4728
	0.552874	20.8391	20.7837	12.4791	5.45465E-01	8.37890E-03				
359.98	14.30610	22.6211	20.7837	12.7726	1.83658E-00	3.01926E-01	19.7385	300	13.2318	13.7858
	8.975310	21.9367	20.7837	12.6601	1.15223E-00	1.89421E-01		500	12.6259	12.6173
	6.206840	21.5813	20.7837	12.6017	7.96818E-01	1.30993E-01		600	12.5661	12.5516
	4.144650	21.3166	20.7837	12.5582	5.32080E-01	8.74715E-02		800	12.5194	12.4991
	2.677940	21.1283	20.7837	12.5272	3.43787E-01	5.65170E-02				
	1.810910	21.0170	20.7837	12.5089	2.32479E-01	3.82186E-02	49.3462	500	12.8586	12.8667
299.99	0.536955	20.8535	20.7837	12.4820	6.89329E-02	1.13323E-02		600	12.7091	12.6698
	11.89980	23.2300	20.7837	12.9296	2.44548E-00	4.58930E-01		800	12.5925	12.5385
	8.937480	22.6212	20.7837	12.8154	1.83670E-00	3.44684E-01	98.6923	400	13.9660	14.6654
	5.010810	21.8143	20.7837	12.6640	1.02975E-00	1.93247E-01		500	13.2464	13.3131
	2.060500	21.2080	20.7837	12.5502	4.23443E-01	7.94654E-02		600	12.9475	12.8798
	0.578633	20.9034	20.7837	12.4930	1.18912E-01	2.23156E-02		800	12.7143	12.6173
260.00	9.070520	23.5244	20.7837	13.0466	2.73987E-00	5.75848E-01				
	5.972270	22.5885	20.7837	12.8499	1.80401E-00	3.79153E-01				
	3.091340	21.7183	20.7837	12.6670	9.33782E-01	1.96256E-01				
	1.020680	21.0928	20.7837	12.5355	3.08309E-01	6.47983E-02				
	0.405823	20.9071	20.7837	12.4965	1.22584E-01	2.57639E-02				
209.97	1.952330	21.8679	20.7837	12.7426	1.08341E-00	2.71932E-01				
	0.800691	21.2288	20.7837	12.5822	4.44329E-00	1.11525E-01				
	0.539946	21.0842	20.7837	12.5459	2.99633E-01	7.52068E-02				
	0.405033	21.0093	20.7837	12.5271	2.24766E-01	5.64154E-02				

^aThe quantities ΔC_P and ΔC_V correspond to $C_P - C_P^0$ and $C_V - C_V^0$, respectively.

Table 12 Specific heat capacities and speed of sound for NF₃

T , K	P , atm	C_P , J/(mol · K)	C_P , J/(mol · K) [44]	C_V , J/(mol · K)	ΔC_P , J/(mol · K)	ΔC_V , J/(mol · K)	u , m · s ⁻¹	u , m · s ⁻¹ (experiment) [41]	u , m · s ⁻¹ [44]
198.98	9.908710	50.2434	42.6793	38.3521	4.517470	0.9399910	152.046	155.028	170.1
	7.948680	49.3498	42.6793	38.1662	3.623880	0.7540530	155.511	158.484	170.1
	5.259310	48.1237	42.6793	37.9111	2.397770	0.4989250	160.139	162.848	170.1
	3.685170	47.4060	42.6793	37.7617	1.680100	0.3495940	162.784	165.244	170.1
	0.986923	46.1759	42.6793	37.5058	0.449947	0.0936246	167.216	169.131	170.1
260.00	9.939300	47.9624	49.4813	37.7916	2.236480	0.3794260	184.706	184.406	191.3
	7.630890	47.4430	49.4813	37.7034	1.717050	0.2913040	186.661	186.027	191.3
	5.257340	46.9089	49.4813	37.6128	1.182970	0.2006960	188.648	187.678	191.3
	3.008140	46.4028	49.4813	37.5270	0.676872	0.1148340	190.508	189.229	191.3
	0.990871	45.9489	49.4813	37.4500	0.222959	0.0378259	192.158	190.623	191.3
300.00	9.928450	47.2851	53.6846	37.6528	1.559120	0.2406360	201.874	199.547	203.9
	7.075250	46.8370	53.6846	37.5836	1.111060	0.1714830	203.446	200.777	203.9
	4.487540	46.4306	53.6846	37.5209	0.704702	0.1087650	204.858	201.902	203.9
	2.364670	46.0973	53.6846	37.4694	0.371336	0.0573126	206.006	202.835	203.9
	0.745127	45.8430	53.6846	37.4302	0.117011	0.0180597	206.876	203.577	203.9
360.00	15.33380	47.2701	59.1447	37.6281	1.544150	0.2159540	222.632	218.252	221.5
	10.47820	46.7811	59.1447	37.5597	1.055180	0.1475700	224.051	219.205	221.5
	4.954350	46.2249	59.1447	37.4819	0.498915	0.0697748	225.646	220.354	221.5
	2.978530	46.0259	59.1447	37.4541	0.299945	0.0419483	226.211	220.786	221.5
	0.993832	45.8260	59.1447	37.4261	0.100081	0.0139967	226.776	221.242	221.5
400.00	15.6062	46.9487	62.212	37.5759	1.222810	0.1638060	236.539	230.667	232.5
	10.2847	46.5318	62.212	37.5201	0.805847	0.1079500	237.505	231.227	232.5
	5.42413	46.1509	62.212	37.4691	0.425002	0.0569328	238.378	231.796	232.5
	3.49766	46.0000	62.212	37.4488	0.274055	0.0367122	238.721	232.040	232.5
	2.11103	45.8913	62.212	37.4343	0.165408	0.0221578	238.967	232.227	232.5

Table 13 Specific heat capacities and speed of sound for BCl_3

T, K	P, atm	$C_P, \text{J}/(\text{mol} \cdot \text{K})$	$C_P, \text{J}/(\text{mol} \cdot \text{K})$ [44]	$C_V, \text{J}/(\text{mol} \cdot \text{K})$	$\Delta C_P,^a \text{J}/(\text{mol} \cdot \text{K})$	$\Delta C_V,^a \text{J}/(\text{mol} \cdot \text{K})$	$u, \text{m} \cdot \text{s}^{-1}$	$u, \text{m} \cdot \text{s}^{-1}$ (experiment) [32]	$u, \text{m} \cdot \text{s}^{-1}$ [44]
290	0.889218	46.5302		37.6295	0.804283	0.2173780	155.996	150.724	
	0.683938	46.3446	61.9478	37.5793	0.618611	0.1671950	156.610	151.568	154.2
	0.495435	46.1741	61.9478	37.5332	0.448113	0.1211140	157.171	152.314	154.2
320	1.487290	46.7276	64.092	37.6613	1.001650	0.2491580	163.111	157.049	161.5
	1.056010	46.4371	64.092	37.5890	0.711194	0.1769080	164.142	158.408	161.5
	0.734271	46.2205	64.092	37.5351	0.494512	0.1230090	164.907	159.393	161.5
	0.476684	46.047	64.092	37.4920	0.321034	0.0798565	165.518	160.162	161.5
360	1.499140	46.4451	66.5643	37.5742	0.719133	0.1621020	174.053	167.603	170.9
	0.997779	46.2046	66.5643	37.5200	0.478632	0.1078900	174.959	168.727	170.9
	0.718480	46.0706	66.5643	37.4898	0.344653	0.0776893	175.462	169.336	170.9
	0.480632	45.9565	66.5643	37.4641	0.230558	0.0519708	175.889	169.846	170.9
400	1.49914	46.2627	68.6616	37.5231	0.536738	0.110986	184.217	178.188	179.7
	1.49618	46.2616	68.6616	37.5229	0.535679	0.110766	184.221	178.324	179.7
	1.49321	46.2606	68.6616	37.5227	0.534615	0.110546	184.225	177.283	179.7
	1.49124	46.2599	68.6616	37.5225	0.53391	0.110401	184.228	177.877	179.7
440	1.536640	46.1515	70.4426	37.4937	0.425555	0.0815864	193.709	186.236	188.2
	0.927708	45.9829	70.4426	37.4614	0.256918	0.0492558	194.383	187.008	188.2
	0.623736	45.8987	70.4426	37.4453	0.172736	0.0331167	194.718	187.387	188.2
	0.515174	45.8686	70.4426	37.4395	0.142671	0.0273527	194.837	187.526	188.2
460	1.539600	46.1050	71.2276	37.4824	0.379108	0.0702370	198.279	190.542	192.2
	1.015540	45.9760	71.2276	37.4585	0.250064	0.0463292	198.797	191.136	192.2
	0.745127	45.9094	71.2276	37.4461	0.183478	0.0339929	199.063	191.434	192.2
	0.522082	45.8545	71.2276	37.4360	0.128556	0.0238175	199.283	191.683	192.2

^aThe quantities ΔC_P and ΔC_V correspond to $C_P - C_P^0$ and $C_V - C_V^0$, respectively.

Table 14 Specific heat capacities and speed of sound for $\text{Ga}(\text{CH}_3)_3$

T, K	P, atm	$C_P, \text{J}/(\text{mol} \cdot \text{K})$	$C_V, \text{J}/(\text{mol} \cdot \text{K})$	$\Delta C_P,^a \text{J}/(\text{mol} \cdot \text{K})$	$\Delta C_V,^a \text{J}/(\text{mol} \cdot \text{K})$	$u, \text{m} \cdot \text{s}^{-1}$	$u, \text{m} \cdot \text{s}^{-1}$ (experiment) [41]
340	1.103380	38.3722	29.3425	9.60115E – 1	2.44125E – 1	174.385	156.998
	0.696768	38.0184	29.2525	6.06298E – 1	1.54161E – 1	175.710	159.196
	0.508265	37.8544	29.2108	4.42271E – 1	1.12455E – 1	176.322	160.165
	0.397730	37.7582	29.1863	3.46088E – 1	8.79985E – 2	176.680	160.726
	0.297064	37.6706	29.1641	2.58493E – 1	6.57259E – 2	177.005	161.238
360	2.174190	39.0115	29.4859	1.59939E – 0	3.87577E – 1	176.921	157.505
	1.748830	38.6986	29.4101	1.28648E – 0	3.11751E – 1	178.148	159.573
	1.313590	38.3784	29.3325	9.66315E – 1	2.34165E – 1	179.397	161.570
	1.115220	38.2325	29.2971	8.20388E – 1	1.98803E – 1	179.963	162.457
	0.904022	38.0772	29.2595	6.65022E – 1	1.61153E – 1	180.565	163.386
	0.811251	38.0089	29.2429	5.96778E – 1	1.44616E – 1	180.829	163.790
400	5.894890	40.6217	29.8107	3.20958E – 0	7.12381E – 1	179.715	155.241
	4.211200	39.7050	29.6072	2.29286E – 0	5.08912E – 1	183.605	161.856
	2.995310	39.0430	29.4603	1.63085E – 0	3.61975E – 1	186.369	166.199
	1.190230	38.0602	29.2422	6.48042E – 1	1.43836E – 1	190.406	172.108
	0.812238	37.8544	29.1965	4.42237E – 1	9.81566E – 2	191.242	173.292
	0.403652	37.6319	29.1471	2.19775E – 1	4.87801E – 2	192.142	174.546
420	8.38391	41.3970	29.9481	3.98490E – 0	8.49726E – 1	180.813	153.941
	6.73378	40.6127	29.7808	3.20058E – 0	6.82481E – 1	184.269	159.972
	4.77079	39.6797	29.5819	2.26757E – 0	4.83528E – 1	188.304	166.424
	2.92919	38.8044	29.3952	1.39225E – 0	2.96879E – 1	192.018	171.951
	2.21071	38.4629	29.3224	1.05076E – 0	2.24060E – 1	193.449	173.996

^aThe quantities ΔC_P and ΔC_V correspond to $C_P - C_P^0$ and $C_V - C_V^0$, respectively.

Table 15 Specific heat capacities and speed of sound for CO₂

T, K	P, atm	$C_p,$ J/(mol · K)	$C_p,$ J/(mol · K) [44]	$C_v,$ J/(mol · K)	$\Delta C_p,$ ^a J/(mol · K)	$\Delta C_v,$ ^a J/(mol · K)	$u, m \cdot s^{-1}$	$u, m \cdot s^{-1}$ (experiment) [38]	$u, m \cdot s^{-1}$ [44]
260.00	8.73348	39.5018	35.3216	29.5038	2.08963E – 0	4.05453E – 1	248.330	247.761	253.5
	6.18574	38.8922	35.3216	29.3855	1.48004E – 0	2.87174E – 1	249.172	246.054	253.5
	4.94755	38.5959	35.3216	29.3280	1.18378E – 0	2.29691E – 1	249.595	247.598	253.5
	3.13220	38.1616	35.3216	29.2437	7.49430E – 1	1.45413E – 1	250.231	249.804	253.5
280.00	9.68981	39.3181	36.2854	29.4476	1.90594E – 0	3.49306E – 1	258.405	252.563	262.0
	6.13333	38.6185	36.2854	29.3194	1.20640E – 0	2.21099E – 1	259.253	256.133	262.0
	4.47323	38.2920	36.2854	29.2596	8.79862E – 1	1.61254E – 1	259.670	257.750	262.0
	2.76644	37.9563	36.2854	29.1981	5.44146E – 1	9.97269E – 2	260.114	259.377	262.0
300.00	10.0403	39.0632	37.2184	29.3859	1.65103E – 0	2.87533E – 1	268.221	262.360	270.1
	7.45611	38.6382	37.2184	29.3119	1.22609E – 0	2.13528E – 1	268.644	264.422	270.1
	4.94301	38.2250	37.2184	29.2399	8.12833E – 1	1.41558E – 1	269.081	266.392	270.1
	2.77177	37.8679	37.2184	29.1777	4.55794E – 1	7.93780E – 2	269.480	268.065	270.1
312.75	9.46834	38.8116	37.7949	29.3349	1.39951E – 0	2.36607E – 1	274.367	268.804	275.2
	6.27910	38.3402	37.7949	29.2552	9.28110E – 1	1.56910E – 1	274.777	271.003	275.2
	4.35904	38.0564	37.7949	29.2073	6.44307E – 1	1.08929E – 1	275.042	272.312	275.2
	3.89045	37.9872	37.7949	29.1955	5.75045E – 1	9.72194E – 2	275.109	272.630	275.2
332.75	9.45621	38.6068	38.6663	29.2919	1.19467E – 0	1.93596E – 1	283.549	277.777	283.0
	6.56028	38.2409	38.6663	29.2326	8.28807E – 1	1.34308E – 1	283.771	279.393	283.0
	5.13851	38.0613	38.6663	29.2035	6.49186E – 1	1.05200E – 1	283.891	580.761	283.0
	3.12144	37.8065	38.6663	29.1622	3.94354E – 1	6.39048E – 2	284.071	281.299	283.0

^aThe quantities ΔC_p and ΔC_v correspond to $C_p - C_p^0$ and $C_v - C_v^0$, respectively.

Table 16 Specific heat capacities and speed of sound for Cl₂

T, K	P, atm	$C_p,$ J/(mol · K)	$C_p,$ J/(mol · K) [44]	$C_v,$ J/(mol · K)	$\Delta C_p,$ ^a J/(mol · K)	$\Delta C_v,$ ^a J/(mol · K)	$u, m \cdot s^{-1}$	$u, m \cdot s^{-1}$ (experiment) [43]	$u, m \cdot s^{-1}$ [44]
260	1.892230	30.5238	33.2053	21.2172	1.42548E – 0	4.32681E – 1	200.481	197.372	201.7
	1.637310	30.3318	33.2053	21.1589	1.23344E – 0	3.74390E – 1	201.312	197.982	201.7
	1.401530	30.1541	33.2053	21.1050	1.05582E – 0	3.20477E – 1	202.081	198.544	201.7
	1.140000	29.9571	33.2053	21.0452	8.58798E – 1	2.60674E – 1	202.933	199.149	201.7
300	6.502050	32.2243	33.9782	21.6259	3.12600E – 0	8.41386E – 1	206.919	205.116	215.8
	3.944340	30.9947	33.9782	21.2949	1.89633E – 0	5.10410E – 1	212.871	209.596	215.8
	2.506000	30.3031	33.9782	21.1088	1.20481E – 0	3.24284E – 1	216.199	211.948	215.8
	1.365110	29.7546	33.9782	20.9612	6.56308E – 1	1.76650E – 1	218.834	213.747	215.8
320	9.857190	33.0008	34.3043	21.7793	3.90250E – 0	9.94732E – 1	209.619	208.547	222.5
	5.965560	31.4601	34.3043	21.3865	2.36178E – 0	6.02010E – 1	217.442	214.510	222.5
	2.959590	30.2700	34.3043	21.0832	1.17171E – 0	2.98665E – 1	223.415	218.711	222.5
	0.982482	29.4873	34.3043	20.8837	3.88968E – 1	9.91466E – 2	227.324	221.313	222.5
360	14.92150	33.2915	34.8574	21.7528	4.19316E – 0	9.68244E – 1	220.689	220.055	235.5
	10.17140	31.9566	34.8574	21.4445	2.85832E – 0	6.60016E – 1	227.950	225.346	235.5
	6.673080	30.9736	34.8574	21.2175	1.87523E – 0	4.33011E – 1	233.223	228.991	235.5
	3.027090	29.9490	34.8574	20.9809	8.50657E – 1	1.96426E – 1	238.672	232.607	235.5
400	1.699380	29.5759	34.8574	20.8948	4.77552E – 1	1.10272E – 1	240.648	233.888	235.5
	14.65090	32.1629	35.297	21.4333	3.06459E – 0	6.48760E – 1	239.345	236.872	247.7
	12.78970	31.7736	35.297	21.3509	2.67529E – 0	5.66346E – 1	241.531	238.334	247.7
	8.278310	30.8299	35.297	21.1511	1.73161E – 0	3.66574E – 1	246.790	241.756	247.7
440	2.053690	29.5279	35.297	20.8755	4.29580E – 1	9.09400E – 2	253.970	246.288	247.7
	1.052450	29.3185	35.297	20.8311	2.20147E – 1	4.66041E – 2	255.119	246.982	247.7
	14.96930	31.5151	35.6515	21.2585	2.41678E – 0	4.73978E – 1	255.211	251.197	259.4
	10.26210	30.7551	35.6515	21.1095	1.65682E – 0	3.24934E – 1	259.530	253.829	259.4
440	7.503380	30.3097	35.6515	21.0221	1.21142E – 0	2.37583E – 1	262.045	255.352	259.4
	5.555290	29.9952	35.6515	20.9604	8.96900E – 1	1.75899E – 1	263.814	256.405	259.4
	4.071160	29.7556	35.6515	20.9134	6.57287E – 1	1.28907E – 1	265.159	257.218	259.4

^aThe quantities ΔC_p and ΔC_v correspond to $C_p - C_p^0$ and $C_v - C_v^0$, respectively.

Table 17 Specific heat capacities for F₂

P , atm	T , K	Density, mol/dm ³ [40]	C_P , J/(mol · K)	C_P , J/(mol · K) [40]	C_P , J/(mol · K) [44]	C_V , J/(mol · K)	C_V , J/(mol · K) [40]	ΔC_P , ^a J/(mol · K)	ΔC_V , ^a J/(mol · K)
0.10000	100	1.124282	29.1684	30.07	32.6348	20.8025	21.10	7.00542E-2	1.79965E-2
	120	0.102716	29.1397	29.66	30.7130	20.7936	20.97	4.13796E-2	9.12690E-3
	140	0.0876736	29.1255	29.51	29.8497	20.7898	20.96	2.71472E-2	5.28753E-3
	160	0.0765307	29.1174	29.51	29.5389	20.7879	21.03	1.91150E-2	3.36405E-3
	180	0.0679263	29.1125	29.62	29.8497	20.7868	21.18	1.41584E-2	2.29475E-3
	200	0.0610741	29.1092	29.81	29.6863	20.7862	21.40	1.08926E-2	1.65209E-3
	220	0.0554849	29.1070	31.07	29.9492	20.7858	21.68	8.63036E-3	1.24166E-3
	240	0.0508373	29.1053	30.37	30.2676	20.7855	22.00	7.00036E-3	9.66439E-4
	260	0.0469108	29.1041	30.70	30.6134	20.7853	22.34	5.78792E-3	7.74353E-4
	300	0.0406384	29.1025	31.41	31.3353	20.7851	23.06	4.13954E-3	5.32775E-4
0.493462	120	0.532031	29.3025	32.17	30.7122	20.8296	21.60	2.04192E-1	4.50378E-2
	140	0.445550	29.2323	30.92	29.8504	20.8106	21.31	1.33961E-1	2.60919E-2
	160	0.384965	29.1927	30.41	29.5381	20.8011	21.23	9.43251E-2	1.66003E-2
	180	0.339591	29.1682	30.25	29.5275	20.7958	21.31	6.98662E-2	1.13237E-2
	200	0.304129	29.1521	30.28	29.6863	20.7927	21.49	5.37507E-2	8.15244E-3
	220	0.275558	29.1409	30.43	29.9492	20.7906	21.75	4.25875E-2	6.1271E-3
	240	0.252004	29.1329	30.67	30.2676	20.7893	22.05	3.45441E-2	4.76900E-3
	260	0.232220	29.1269	30.95	30.6134	20.7883	22.40	2.85612E-2	3.82113E-3
	300	0.200834	29.1188	31.60	31.3353	20.7871	23.12	2.04270E-2	2.62904E-3
0.986923	120	1.14361	29.5067	36.87	30.7122	20.8746	22.90	4.08385E-1	9.00755E-2
	140	0.927643	29.3662	33.07	29.8504	20.8367	21.82	2.67922E-1	5.21839E-2
	160	0.789600	29.2870	31.68	29.5381	20.8177	21.49	1.88650E-1	3.32006E-2
	180	0.690680	29.2381	31.10	29.5275	20.8072	21.46	1.39732E-1	2.26474E-2
	200	0.615315	29.2058	30.90	29.6863	20.8008	21.60	1.07501E-1	1.63049E-2
	220	0.555576	29.1835	30.91	29.9484	20.7968	21.83	8.51750E-2	1.22542E-2
	240	0.506870	29.1674	31.05	30.2676	20.7941	22.13	6.90882E-2	9.53801E-3
	260	0.466297	29.1554	31.27	30.6134	20.7922	22.46	5.71223E-2	7.64227E-3
	300	0.402396	29.1392	31.84	31.3353	20.7898	23.19	4.08541E-2	5.25808E-3
1.38169	120	1.71988	29.6701	42.98	30.7122	20.9106	24.26	5.71739E-1	1.26106E-1
	140	1.34546	29.4734	35.21	29.8504	20.8576	22.28	3.75091E-1	7.30574E-2
	160	1.12914	29.3624	32.82	29.5381	20.8310	21.71	2.6411E-1	4.64808E-2
	180	0.980376	29.2940	31.83	29.5275	20.8162	21.59	1.95625E-1	3.17064E-2
	200	0.869520	29.2488	31.42	29.6863	20.8073	21.69	1.50502E-1	2.28268E-2
	220	0.782845	29.2176	31.30	29.9484	20.8017	21.90	1.19245E-1	1.71559E-2
	240	0.712816	29.1950	31.36	30.2676	20.7979	22.19	9.67234E-2	1.33532E-2
	260	0.654856	29.1783	31.53	30.6134	20.7952	22.52	7.99713E-2	1.06992E-2
	300	0.564142	29.1555	32.02	32.8200	20.7919	23.24	5.71957E-2	7.36132E-3

^aThe quantities ΔC_P and ΔC_V correspond to $C_P - C_P^0$ and $C_V - C_V^0$, respectively.

Table 18 Specific heat capacities for H₂

P , atm	T , K	C_P , J/(mol · K)	C_P , J/(mol · K) [48]	C_P , J/(mol · K) [44]	C_V , J/(mol · K)	C_V , J/(mol · K) [48]	ΔC_P , ^a J/(mol · K)	ΔC_V , ^a J/(mol · K)
0.394769	100	29.1300	27.02		20.7886	18.69	3.16810E-2	4.06748E-3
	150	29.1109	32.68		20.7862	24.35	1.25580E-2	1.66651E-3
	200	29.1049	32.40		20.7855	24.08	6.57919E-3	1.00730E-3
	250	29.1023	30.88		20.7852	22.57	3.98342E-3	7.20769E-4
	300	29.1010	29.93		20.7851	21.61	2.63497E-3	5.62537E-4
	360	29.1001	29.45		20.7850	21.14	1.73314E-3	4.46112E-4
	400	29.0997	29.33		20.7849	21.02	1.35536E-3	3.92196E-4
1.000000	100	29.1786	27.02	26.4233	20.7948	18.69	8.02536E-2	1.03034E-2
	150	29.1301	32.69	26.1001	20.7887	24.35	3.18123E-2	4.22148E-3
	200	29.1150	32.40	27.4466	20.7871	24.08	1.66659E-2	2.55162E-3
	250	29.1084	30.89	28.3583	20.7863	22.57	1.00905E-2	1.82580E-3
	300	29.1050	29.93	28.8508	20.7859	21.61	6.67472E-3	1.42498E-3
	360	29.1027	29.46	29.1165	20.7856	21.14	4.39025E-3	1.13006E-3
	400	29.1018	29.33	29.1907	20.7855	21.02	3.43328E-3	9.9348E-4
9.86923	100	29.8904	27.67	26.4233	20.8862	18.76	7.92041E-1	0.101687000
	150	29.4123	32.94	26.1001	20.8262	24.40	3.13963E-1	0.041662700
	200	29.2628	32.54	27.4466	20.8097	24.11	1.64480E-1	0.025182500
	250	29.1979	30.97	28.3583	20.8025	22.59	9.95856E-2	0.018019200
	300	29.1642	29.98	28.8508	20.7986	21.63	6.58744E-2	0.014063400
	360	29.1417	29.49	29.1165	20.7957	21.15	4.33284E-2	0.011152800
	400	29.1322	29.36	29.1907	20.7943	21.03	3.38839E-2	0.009804890
14.8038	100	30.2864	28.00	26.4233	20.9370	18.79	1.18806E-0	1.52531E-1
	150	29.5693	33.08	26.1001	20.8470	24.43	4.70945E-1	6.24941E-2
	200	29.3450	32.61	27.4466	20.8223	24.13	2.46719E-1	3.77737E-2
	250	29.2477	31.01	28.3583	20.8115	22.61	1.49378E-1	2.70288E-2
	300	29.1971	30.01	28.8508	20.8056	21.64	9.88115E-2	2.10951E-2
	360	29.1633	29.50	29.1165	20.8012	21.16	6.49926E-2	1.67292E-2
	400	29.1492	29.37	29.1907	20.7992	21.04	5.08258E-2	1.47073E-2
23.6862	100	30.9992	28.57	26.4233	21.0286	18.85	1.90090E-0	2.44049E-1
	150	29.8518	33.32	26.1001	20.8845	24.47	7.53512E-1	9.99906E-2
	200	29.4931	32.73	27.4466	20.8450	24.16	3.94751E-1	6.04380E-2
	250	29.3373	31.08	28.3583	20.8278	22.63	2.39005E-1	4.32461E-2
	300	29.2564	30.06	28.8508	20.8183	21.66	1.58098E-1	3.37522E-2
	360	29.2023	29.53	29.1165	20.8113	21.18	1.03988E-1	2.67667E-2
	400	29.1796	29.39	29.1907	20.8081	21.05	8.13213E-1	2.35317E-2

^aThe quantities ΔC_P and ΔC_V correspond to $C_P - C_P^0$ and $C_V - C_V^0$, respectively.

Table 19 Specific heat capacities and speed of sound for air

P , atm	T , K	Density, mol/dm ³ [37]	C_p , J/(mol · K)	C_p , J/(mol · K) [37]	C_v , J/(mol · K)	C_v , J/(mol · K) [37]	ΔC_p , ^a J/(mol · K)	ΔC_v , ^a J/(mol · K)	u , m · s ⁻¹	u , m · s ⁻¹ [37]
1.000000	100.0	0.124490	29.7596	30.13	20.9344	21.09	6.61290E-1	1.49919E-1	200.120	198.2
	200.0	0.061079	29.2087	29.16	20.8000	20.74	1.10329E-1	1.54602E-2	283.847	283.4
	300.0	0.040634	29.1410	29.15	20.7899	20.80	4.27181E-2	5.41928E-3	347.659	347.4
	400.0	0.030461	29.1206	29.38	20.7875	21.04	2.22346E-2	2.96845E-3	401.390	400.5
	500.0	0.024365	29.1118	29.83	20.7865	21.50	1.34660E-2	2.00202E-3	448.712	446.4
	600.0	0.020303	29.1073	30.45	20.7860	22.13	8.94176E-3	1.50734E-3	491.489	487.1
	800.0	0.015228	29.1030	31.82	20.7855	23.51	4.66294E-3	1.01269E-3	567.440	557.8
	1000	0.012183	29.1011	33.05	20.7853	24.73	2.78689E-3	7.64832E-4	634.356	619.6
	2000	0.006092	29.0988	36.21	20.7849	27.90	4.89294E-4	3.41278E-4	896.929	863.5
4.93462	100.0	0.67871	32.3615	36.53	21.5243	23.07	3.26321E-0	7.39792E-1	198.607	187.4
	200.0	0.30425	29.6428	29.68	20.8608	20.84	5.44432E-1	7.62903E-2	284.939	282.6
	300.0	0.20075	29.3091	29.33	20.8113	20.82	2.10797E-1	2.67421E-2	349.034	347.8
	400.0	0.15020	29.2080	29.47	20.7992	21.06	1.09719E-1	1.46482E-2	402.763	401.3
	500.0	0.12007	29.1648	29.89	20.7944	21.51	6.64497E-2	9.87918E-3	450.023	447.3
	600.0	0.10004	29.1425	30.48	20.7920	22.13	4.41241E-2	7.43816E-3	492.727	488.0
	800.0	0.075038	29.1213	31.84	20.7895	23.51	2.30098E-2	4.99725E-3	568.544	558.7
	1000	0.060042	29.1121	33.06	20.7883	24.74	1.37522E-2	3.77415E-3	635.350	620.4
	2000	0.030042	29.1007	36.21	20.7862	27.90	2.41448E-3	1.68408E-3	897.607	864.1
9.86923	150.0	0.85577	31.3100	32.30	21.1511	21.42	2.21169E-0	3.66589E-1	247.021	238.5
	200.0	0.61574	30.1872	30.36	20.9371	20.95	1.08886E-0	1.52581E-1	286.392	281.7
	300.0	0.40205	29.5199	29.56	20.8380	20.86	4.21595E-1	5.34842E-2	350.779	348.4
	400.0	0.30009	29.3178	29.58	20.8138	21.07	2.19439E-1	2.92964E-2	404.490	402.3
	500.0	0.23974	29.2312	29.95	20.8043	21.53	1.32899E-1	1.97584E-2	451.668	448.5
	600.0	0.19972	29.1866	30.53	20.7994	22.14	8.82483E-2	1.48763E-2	494.280	489.2
	800.0	0.14981	29.1443	31.87	20.7945	23.52	4.60197E-2	9.99450E-3	569.927	559.9
	1000	0.11990	29.1258	33.07	20.7921	24.74	2.75045E-2	7.54831E-3	636.595	621.5
	2000	0.060031	29.1032	36.22	20.7879	27.90	4.82896E-3	3.36815E-3	898.457	864.9
19.7385	150.0	1.8462	33.5217	37.01	21.5177	22.22	4.42337E-0	7.33178E-1	249.121	231.3
	200.0	1.2610	31.2761	31.84	21.0897	21.19	2.17773E-0 q	3.05161E-1	289.559	280.3
	300.0	0.80597	29.9415	30.03	20.8915	20.93	8.43189E-1	1.06968E-1	354.341	349.9
	400.0	0.59886	29.5372	29.81	20.8431	21.11	4.38877E-1	5.85927E-2	407.966	404.5
	500.0	0.47783	29.3641	30.09	20.8240	21.55	2.65799E-1	3.95167E-2	454.962	450.8
	600.0	0.39794	29.2748	30.62	20.8143	22.16	1.76497E-1	2.97526E-2	497.383	491.5
	800.0	0.29857	29.1904	31.91	20.8045	23.53	9.20393E-2	1.99890E-2	572.688	562.1
	1000	0.23905	29.1533	33.10	20.7996	24.75	5.50090E-2	1.50966E-2	639.081	623.6
	2000	0.1185	29.1080	36.22	20.7913	27.90	9.65791E-3	6.73630E-3	900.153	866.4

^aThe quantities ΔC_p and ΔC_v correspond to $C_p - C_p^0$ and $C_v - C_v^0$, respectively.

used to calculate the speed of sound. From the results, the proposed method produced good results in a wide range of parameters. Therefore, the results of our calculations, in this study, will be useful for different aspects of technology and industry.

One of the advances of this study is the use of series formulas of the exponential functions for the established formulas of second virial coefficients. It is well known that the series formulas of the exponential functions are general and provide accurate results of arbitrary values of parameters. Therefore, the obtained formulas for the second virial coefficient and related quantities are valid in the arbitrary values of parameters. Note that the virial coefficients are applied at the specific pressure and temperature ranges that molecules show the gas characteristic. At low temperatures and high pressures, the gases begin to form liquids with great deviations from ideal behavior. As seen from the tables, at low temperatures and high pressures, the calculation results obtained from the second virial coefficient deviate.

IV. Conclusions

In this paper, an efficient approach is described for the analytical evaluation of the specific heat capacity and speed of sound, through use of the virial coefficient. Results for the specific heat capacity and speed of sound, for several different molecules, are reported and then compared with data from the literature.

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